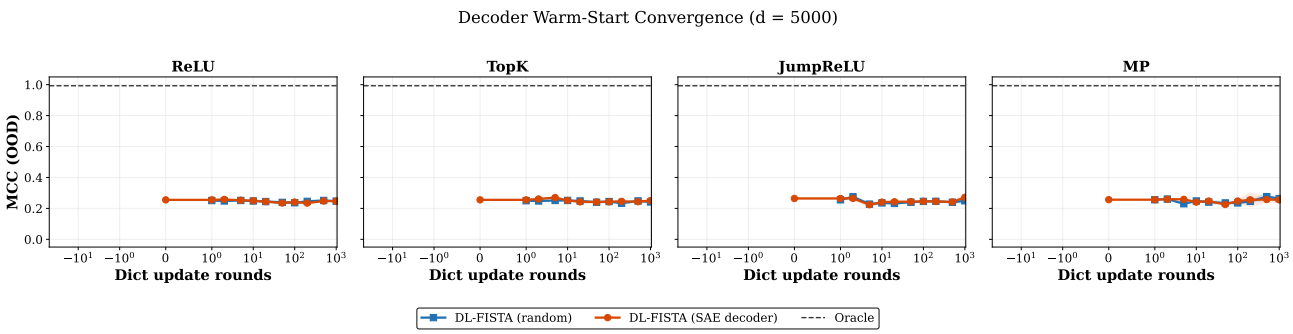

Figure 41: Decoder warm-start convergence, $d_z = 5000$.

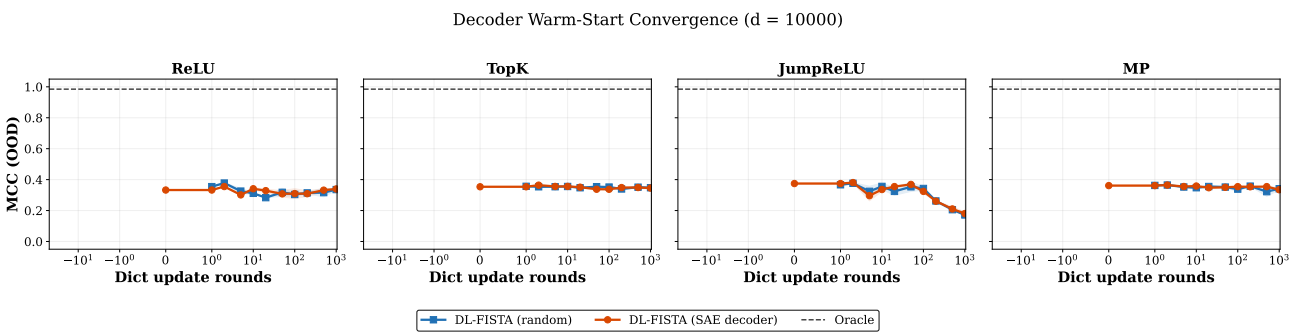

Figure 42: Decoder warm-start convergence, $d_z = 10000$.

E THEORETICAL MODEL FOR TOY SETTING

We study the geometry of a system where a sparse source vector $z \in [0,1]^3$ with at most two non-zero elements ($\|z\|_0 \leq 2$) is linearly projected to an observation $y \in \mathbb{R}^2$ (see Fig. 1):

$$y = Az. \tag{11}$$

The sparsity constraint implies that any observation is a combination of at most two active source components. Whenever active, we assume that each source follows a uniform distribution $z_i \mid i \text{ active} \sim \text{Uniform}(0,1)$. The training data is considered *independent and identically distributed* (IID) and is generated from combinations of sources (z_1, z_2) or (z_2, z_3). The test data is considered *out-of-distribution* (OOD) and is generated from the novel combination (z_1, z_3). Our goal is to determine whether the first variable z_1 is above a certain, safety-relevant, threshold $z_1 = \frac{1}{2}$.

To analyze the geometry, we examine the columns $A_i \in \mathbb{R}^2$ of the projection matrix. We define the angles $\phi := \angle(A_1, A_2)$ and $\theta := \angle(A_1, A_3)$, which fully determine the system. To simplify the analysis, we make two assumptions:

1. We align our coordinate system and fix the magnitude of the first basis vector relative to our threshold, such that $A_1 = (2, 0)$ and $\|A_2\| = \|A_3\| = 1$.
2. To ensure the cones spanned by the vectors do not overlap, we require that $0 < \phi, \theta < \pi$ and $\phi + \theta > \pi$. This is an illustrative way of understanding why and when compressed sensing is possible in this system.

A perfect linear classifier trained on the IID data must separate the space based on the condition $z_1 = \frac{1}{2}$. In the observation space, this corresponds to a line parallel to A_2 and passing through the point $\frac{1}{2}A_1$. This decision boundary is the line parameterized by:

$$y(\beta) = \frac{1}{2}A_1 + \beta A_2, \quad \beta \in \mathbb{R}. \tag{12}$$

The question we are interested in is: *what is the accuracy of this classifier on the OOD data?* We derive the analytically predicted OOD accuracy for this perfect linear IID classifier, separating two cases, in Appendix E.1.

The simulations and analytical prediction are tested and illustrated in Fig. 63 confirming the validity of the theory.

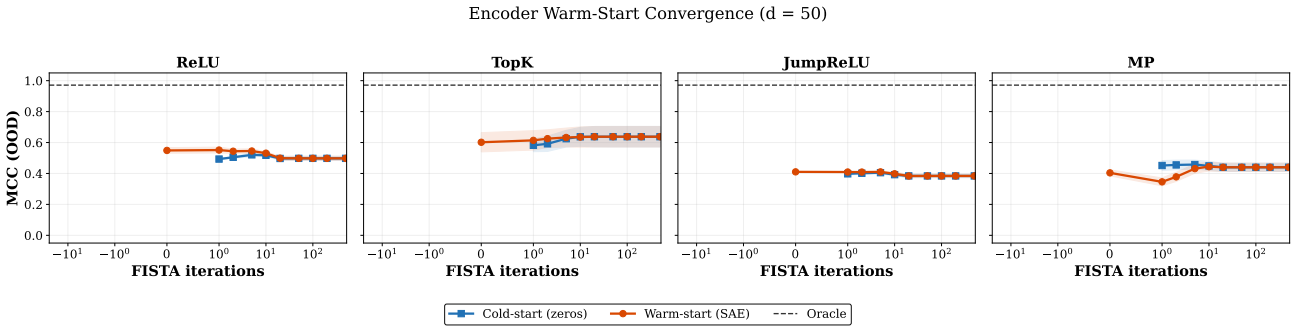

Figure 43: Encoder warm-start convergence, $d_z = 50$.

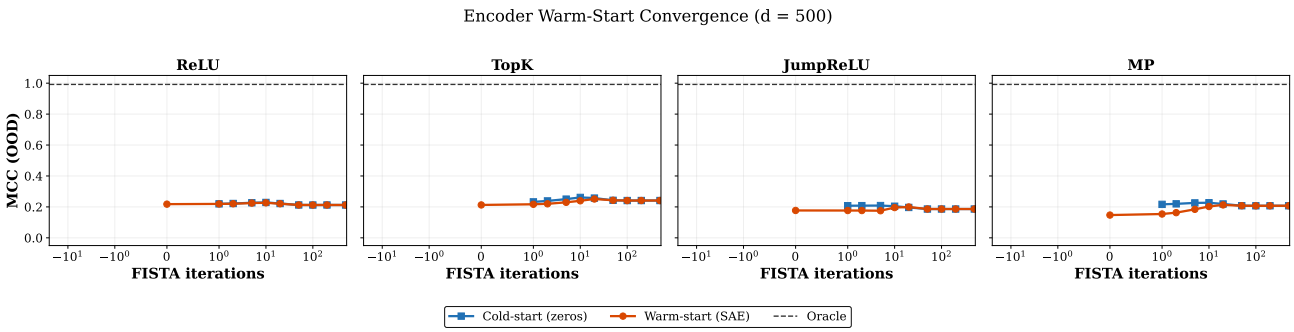

Figure 44: Encoder warm-start convergence, $d_z = 500$.

E.1 DERIVATION

We study the geometry of a system where a sparse source vector $z \in [0,1]^3$ with at most two non-zero elements ($\|z\|_0 \leq 2$) is linearly projected to an observation $y \in \mathbb{R}^2$ (see Fig. 1):

$$y = Az. \tag{13}$$

The sparsity constraint implies that any observation is a combination of at most two active source components. Whenever active, we assume that each source follows a uniform distribution $z_i \mid i$ active $\sim \text{Uniform}(0,1)$. The training data is considered *independent and identically distributed* (IID) and is generated from combinations of sources (z_1, z_2) or (z_2, z_3). The test data is considered *out-of-distribution* (OOD) and is generated from the novel combination (z_1, z_3). Our goal is to determine whether the first variable z_1 is above a certain, safety-relevant, threshold $z_1 = \frac{1}{2}$.

To analyze the geometry, we examine the columns $A_i \in \mathbb{R}^2$ of the projection matrix. We define the angles $\phi := \angle(A_1, A_2)$ and $\theta := \angle(A_1, A_3)$, which fully determine the system. To simplify the analysis, we make two assumptions:

1. We align our coordinate system and fix the magnitude of the first basis vector relative to our threshold, such that $A_1 = (2, 0)$ and $\|A_2\| = \|A_3\| = 1$.
2. To ensure the cones spanned by the vectors do not overlap, we require that $0 < \phi, \theta < \pi$ and $\phi + \theta > \pi$. This is an illustrative way of understanding why and when compressed sensing is possible in this system.

A perfect linear classifier trained on the IID data must separate the space based on the condition $z_1 = \frac{1}{2}$. In the observation space, this corresponds to a line parallel to A_2 and passing through the point $\frac{1}{2}A_1$. This decision boundary is the line parameterized by:

$$y(\beta) = \frac{1}{2}A_1 + \beta A_2, \quad \beta \in \mathbb{R}. \tag{14}$$

The question we are interested in is: *what is the accuracy of this classifier on the OOD data?* Clearly, the perfect linear classifier for the OOD data would have a decision boundary that is parallel to A_3 and shifted by $\frac{1}{2}A_1$, i.e., the line:

$$y_{\text{OOD}}(\beta) = \frac{1}{2}A_1 + \beta A_3, \quad \beta \in \mathbb{R}. \tag{15}$$

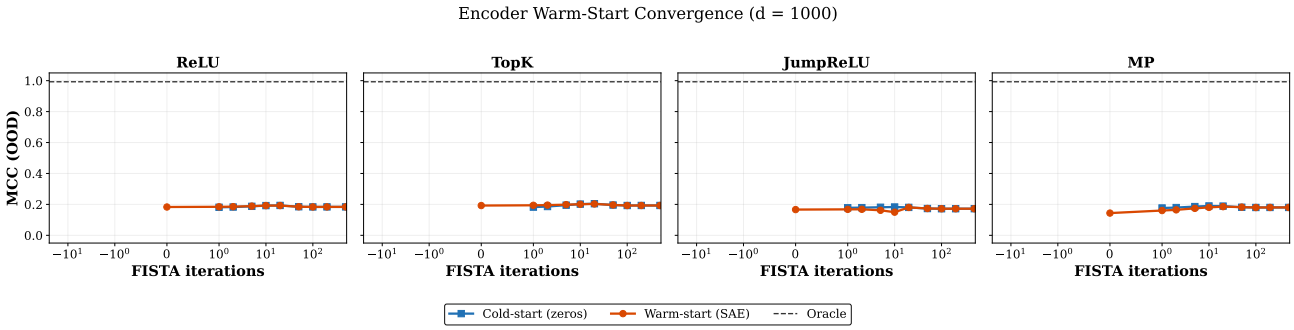

Figure 45: Encoder warm-start convergence, $d_z = 1000$.

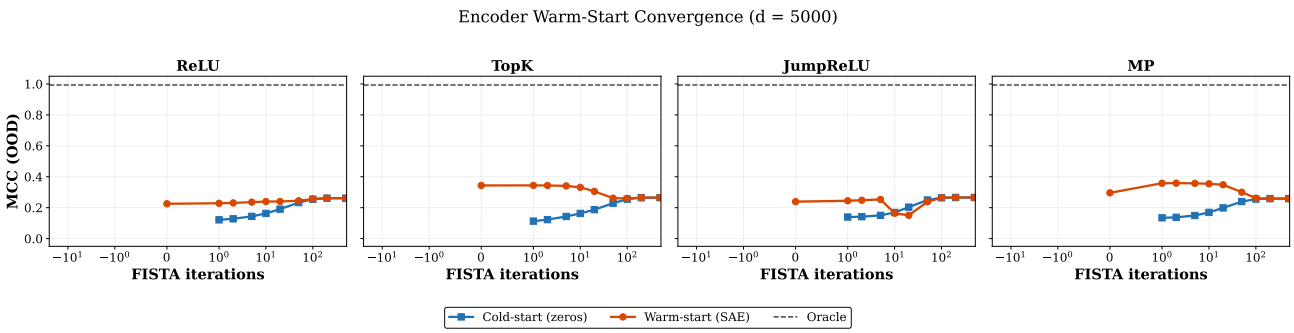

Figure 46: Encoder warm-start convergence, $d_z = 5000$.

Since $\phi + \theta > \pi$, we know that the IID classifier's boundary cannot be aligned with the ideal OOD classifier's boundary, so there must be some OOD error. Moreover, we know that the IID classifier can never 'under-shoot' on the OOD data (that would require $\phi + \theta < \pi$). Consequently, we will only observe *false negatives*—that is, test points with $z_1 > \frac{1}{2}$ that are erroneously classified as safe.

We now have to distinguish: **Case 1**, where the classifier passes right from the top right corner $(A_1 + A_3)$ (Fig. 63 Point C), and **Case 2**, where the classifier passes left from the top right corner $(A_1 + A_3)$ (Fig. 63 Point A).

The separation happens when the classifier passes through the top right corner. In that case it will form a triangle through the points $(\frac{1}{2}A_1, A_1, A_1 + A_3)$, with associated angles $(a, b, c) := (\pi - \phi, \pi - \theta, \phi + \theta - \pi)$. By assumption, the base of this triangle has length 1. Consequently, trigonometry tells us that the first angle must have a fixed relation to the second angle $a = \frac{\pi - b}{2}$. From this it follows that the condition for Case 1 is

$$\frac{\pi - (\pi - \theta)}{2} < \pi - \phi \quad \Rightarrow \quad \phi + \frac{\theta}{2} < \pi. \tag{16}$$

The total area of the right parallelogram $(\frac{1}{2}A_1, A_1, A_1 + A_3, \frac{1}{2}A_1 + A_3)$ is $\alpha = \sin(\theta)$. To compute the area of a triangle within this diagram, we use the fact that the area of a triangle can be computed from one side and the adjacent angles. We will always pick a side with length 1, so that if the adjacent angles are (a, b), the area equals

$$\alpha(a, b) = \frac{\sin(a)\sin(b)}{2\sin(a + b)}. \tag{17}$$

In Case 1, we compute the area (α_1) of the triangle between the classifier and A_1. The base between $\frac{1}{2}A_1$ and A_1 has length 1. The angle on the left is $a_1 = \pi - \phi$ and the angle on the right is $b_1 = \pi - \theta$. Thus, using equation 17, the area is

$$\alpha_1 = \frac{\sin(\pi - \phi)\sin(\pi - \theta)}{2\sin(\phi + \theta - \pi)} = \frac{\sin(\phi)\sin(\theta)}{2\sin(\phi + \theta - \pi)} \tag{18}$$

The OOD accuracy will be 50% for the true negatives, plus 50% times the proportion that the area occupies in the right parallelogram (α)

$$acc_1(\text{OOD}) = \frac{1}{2} + \frac{\alpha_1}{2\alpha}. \tag{19}$$